Online Appendix: Nonconvex Optimization with Spectral Radius Regularization

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Appendix A
Hessian-Vector Operations Derivation

The forward computation for each layer of a network with input x, output y, weights w, activation σ, bias I, error or loss measure $E = E(y)$, and direct derivative $e_k = dE/dy_k$ is given by:

$$x_k = x_k(y_{k-1}, w_k) = \sum_j w_{jk} y_{j(k-1)}$$

$$y_k = y_k(x_k, I_k) = \sigma_k(x_k) + I_k.$$

The backward computation is

$$\frac{\partial E}{\partial y_k} = e_k(y_k) + \sum_j w_{jk} \frac{\partial E}{\partial x_j},$$

$$\frac{\partial E}{\partial x_k} = \sigma_k'(x_k) \frac{\partial E}{\partial y_k},$$

$$\frac{\partial E}{\partial w_{jk}} = y_k \frac{\partial E}{\partial x_j}.$$

Applying $\mathcal{R}_v\{\cdot\}$ to the forward pass gives

$$\mathcal{R}_v\{x_k; w\} = \sum_j \left(w_{jk} \mathcal{R}_v\{y_{j(k-1)}; w\} + v_{jk} y_{j(k-1)} \right),$$

$$\mathcal{R}_v\{y_k; w\} = \mathcal{R}_v\{x_k; w\} \sigma_k'(x_k).$$

The backward computation follows as

$$\mathcal{R}_v\left\{\frac{\partial E}{\partial y_k}; w\right\} = e_k'(y_k)\mathcal{R}_v\{y_k; w\} + \sum_j \left[w_{jk} \mathcal{R}_v\left\{\frac{\partial E}{\partial x_j}; w\right\} + v_{jk}\frac{\partial E}{\partial x_j}\right]$$

$$\mathcal{R}_v\left\{\frac{\partial E}{\partial x_k}; w\right\} = \sigma_k'(x_k)\mathcal{R}_v\left\{\frac{\partial E}{\partial y_k}; w\right\} + \mathcal{R}_v\{x_k; w\}\sigma_k''(x_k)\frac{\partial E}{\partial y_k}$$

$$\mathcal{R}_v\left\{\frac{\partial E}{\partial w_{jk}}; w\right\} = y_k \mathcal{R}_v\left\{\frac{\partial E}{\partial x_j}; w\right\} + \mathcal{R}_v\{y_k; w\}\frac{\partial E}{\partial x_j}.$$

This yields the result found in Pearlmutter [1]. However, we extend it one step further by applying $\mathcal{R}_v^2\{\cdot\} = \mathcal{R}_v\{\mathcal{R}_v\{\cdot\}\}$ to the original forward pass:

$$\mathcal{R}_v^2\{x_k; w\} = \sum_j \left[w_{jk}\mathcal{R}_v^2\{y_{j(k-1)}; w\} + 2v_{jk}\mathcal{R}_v\{y_{j(k-1)}; w\}\right]$$

$$\mathcal{R}_v^2\{y_k; w\} = \mathcal{R}_v^2\{x_k; w\}\sigma_k'(x_k) + (\mathcal{R}_v\{x_k; w\})^2\sigma_k''(x_k).$$

The backward computation follows as

$$\mathcal{R}_v^2\left\{\frac{\partial E}{\partial y_k}; w\right\} = e_k''(y_k)(\mathcal{R}_v\{y_k; w\})^2 + e_k'(y_k)\mathcal{R}_v^2\{y_k; w\} + \sum_j \left[w_{jk}\mathcal{R}_v^2\left\{\frac{\partial E}{\partial x_j}; w\right\} + 2v_{jk}\mathcal{R}_v\left\{\frac{\partial E}{\partial x_j}; w\right\}\right]$$

$$\mathcal{R}_v^2\left\{\frac{\partial E}{\partial x_k}; w\right\} = 2\mathcal{R}_v\{x_k; w\}\sigma_k''(x_k)\mathcal{R}_v\left\{\frac{\partial E}{\partial y_k}; w\right\} + \sigma_k'(x_k)\mathcal{R}_v^2\left\{\frac{\partial E}{\partial y_k}; w\right\} + \mathcal{R}_v^2\{x_k; w\}\sigma_k''(x_k)\frac{\partial E}{\partial y_k} + (\mathcal{R}_v\{x_k; w\})^2\sigma_k'''(x_k)\frac{\partial E}{\partial y_k}$$

$$\mathcal{R}_v^2\left\{\frac{\partial E}{\partial w_{jk}}; w\right\} = 2\mathcal{R}_v\{y_k; w\}\mathcal{R}_v\left\{\frac{\partial E}{\partial x_j}; w\right\} + y_k\mathcal{R}_v^2\left\{\frac{\partial E}{\partial x_j}; w\right\} + \mathcal{R}_v^2\{y_k; w\}\frac{\partial E}{\partial x_j}$$

The original formulation $\mathcal{R}_v\{\cdot\}$ allows us to efficiently compute $H(w)v$, which can be used to compute $\rho(w)$ and/or estimate the eigenvector $\bar{v}$ corresponding to the spectral radius (via power iteration or LOBPCG). However, the extended formulation $\mathcal{R}_v^2\{\cdot\}$ allows us to efficiently compute $v^T \nabla H(w) v$ and thus $\nabla\rho(w)$. This enables us to efficiently compute the gradient of our optimization problem for use in gradient descent methods.

Appendix B
Convergence Analysis Proofs

A. Stochastic Gradient Descent Convergence

First, we prove the gradient-convergence lemma stated in the main paper.

Proof. We start by splitting $v_k^T \nabla H_k v_k$ into its components:

$$v_k^T \nabla H_k v_k = (v_k - \bar{v}_k + \bar{v}_k)^T \nabla H_k (v_k - \bar{v}_k + \bar{v}_k).$$

$$= (v_k - \bar{v}_k)^T \nabla H_k (v_k - \bar{v}_k) + 2(v_k - \bar{v}_k)^T \nabla H_k \bar{v}_k + \bar{v}_k^T \nabla H_k \bar{v}_k.$$

The last term, $\bar{v}_k^T \nabla H_k \bar{v}_k = \nabla \bar{\rho}_k$, by definition. We apply the triangle inequality and bound the other terms. Since $\bar{v}_k$ is a unit eigenvector, $\|\bar{v}_k\| = 1$ and its norm drops from the first bound below. Given the convergence criterion on v_k and Assumptions A1 and A4 (with $\|H(w) - H(\omega)\| \leq L\|w - \omega\|$ for all $w, \omega \in \mathbb{R}^n$ and $L \geq 0$), it follows that

$$\|(v_k - \bar{v}_k)^T \nabla H_k \bar{v}_k\| \leq L\|v_k - \bar{v}_k\| \leq L\epsilon_k.$$

For the first term, we similarly obtain

$$\|(v_k - \bar{v}_k)^T \nabla H_k (v_k - \bar{v}_k)\| \leq L\|v_k - \bar{v}_k\|^2 \leq L\epsilon_k^2.$$

Given Assumption A5,

$$\lim_{k\to\infty} v_k^T \nabla H_k v_k = \lim_{k\to\infty} \bar{v}_k^T \nabla H_k \bar{v}_k = \lim_{k\to\infty} \nabla\bar{\rho}_k.$$

□

Next, we prove that the power-iteration and LOBPCG algorithms satisfy the update-moment lemma stated in the main paper.

Proof. We begin by splitting p_k into its components:

$$\|p_k\|^2 = \|p_k - \nabla g_k + \nabla g_k\|^2 = \|p_k - \nabla g_k\|^2 + \|\nabla g_k\|^2 + 2(p_k - \nabla g_k)^T \nabla g_k.$$

Let us first assume $\rho_k > K$. By the definition of g_k and the triangle inequality,

$$\|\nabla g_k\|^2 \leq \|\nabla f_k\|^2 + 2\mu\|\nabla f_k\|\|\nabla \rho_k\| + \mu^2\|\nabla \rho_k\|^2.$$

Taking the expectation (with respect to batch B_k conditioned on the history $\mathcal{P}_k$) and applying the Cauchy–Schwarz inequality yields

$$\mathbb{E}_{B_k}[\|p_k\|^2 \mid \mathcal{P}_k] \leq \mathbb{E}_{B_k}[\|\nabla f_k\|^2 \mid \mathcal{P}_k] + 2\mu(\mathbb{E}_{B_k}[\|\nabla f_k\|^2 \mid \mathcal{P}_k])^{1/2}(\mathbb{E}_{B_k}[\|\nabla \rho_k\|^2 \mid \mathcal{P}_k])^{1/2} + \mu^2\mathbb{E}_{B_k}[\|\nabla \rho_k\|^2 \mid \mathcal{P}_k].$$

Applying Hölder's inequality with $\|\nabla f_k\|^2$, $\|\nabla \rho_k\|$, $p = 3/2$, and $q = 3$ yields

$$\mathbb{E}_{B_k}[\|\nabla f_k\|^2\|\nabla \rho_k\| \mid \mathcal{P}_k] \leq (\mathbb{E}_{B_k}[\|\nabla f_k\|^3 \mid \mathcal{P}_k])^{2/3}(\mathbb{E}_{B_k}[\|\nabla \rho_k\|^3 \mid \mathcal{P}_k])^{1/3}.$$

Similarly, using $\|\nabla f_k\|^3$, $\|\nabla \rho_k\|$, $p = 4/3$, and $q = 4$ yields

$$\mathbb{E}_{B_k}[\|\nabla f_k\|^3\|\nabla \rho_k\| \mid \mathcal{P}_k] \leq (\mathbb{E}_{B_k}[\|\nabla f_k\|^4 \mid \mathcal{P}_k])^{3/4}.$$

Applying this, we obtain bounds for higher moments as required by Assumption A3. This implies that $\mathbb{E}_{B_k}[\|p_k\|^j \mid \mathcal{P}_k] \leq \bar{A}_j + \bar{B}_j\|w_k\|^j$ for $j = 2, 3, 4$, some positive constants $\bar{A}_j, \bar{B}_j$, and any k. Combining this with the preceding results shows that the second, third, and fourth moments of the update term are bounded, as required. If $\rho_k \leq K$, then the penalty is inactive and $p_k = \nabla f_k$; the statement follows from Assumption A3. $\square$

The rest of this proof uses our assumptions and lemmas and follows Bottou's [2] proof that SGD converges. In the next step, we prove the bounded-iterates lemma stated in the main paper.

Proof. Let $\varphi(x)$ be defined to confine the iterates. Taking the expectation, we have

$$\mathbb{E}_{B_k}[\psi_{k+1} - \psi_k \mid \mathcal{P}_k] \leq -2\alpha_k w_k^T \nabla g_k \psi'(\|w_k\|^2) + \alpha_k^2 \mathbb{E}_{B_k}[\|p_k\|^2 \mid \mathcal{P}_k]\psi'(\|w_k\|^2) + \dots.$$

Given Assumption A2, for sufficiently large k, $\alpha_k^2 \geq \alpha_k^3 \geq \alpha_k^4$. Due to the update-moment lemma, there exist positive constants A_0, B_0 such that the inequality is bounded by $\alpha_k^2(A + B\psi_k)$. We then transform the expectation inequality and use the Quasi-Martingale Convergence Theorem. Since the involved series almost surely converge, we combine the terms to obtain that the iterates must be bounded. $\square$

Finally, we prove the almost-sure convergence theorem stated in the main paper.

Proof. All statements here are taken almost surely. By Assumption A1, we have $f \in \mathcal{C}^5$. From linear algebra, the Hessian of $\rho(w)$ is continuous. This implies that $g \in \mathcal{C}^3$, and thus, by the bounded-iterates lemma, it is bounded on the set of all iterates. We can bound differences in the loss criteria g_k using a first-order Taylor expansion and bounding the second derivatives. Taking the expectation, we obtain

$$\mathbb{E}_{B_k}[g_{k+1} - g_k \mid \mathcal{P}_k] \leq -\alpha_k \mathbb{E}_{B_k}[p_k^T \nabla g_k \mid \mathcal{P}_k] + \alpha_k^2 \mathbb{E}_{B_k}[\|p_k\|^2 \mid \mathcal{P}_k]K_1.$$

Applying the Cauchy–Schwarz inequality and bounding the error term gives

$$\mathbb{E}_{B_k}[g_{k+1} - g_k \mid \mathcal{P}_k] \leq \alpha_k^2 K_1 K_2 + \alpha_k \epsilon_k K_3.$$

By the Quasi-Martingale Convergence Theorem, g_k converges almost surely. Since g_k converges, $\sum_{k=1}^{\infty} \alpha_k \|\nabla g_k\|^2 < \infty$. We define $\theta_k = \|\nabla g_k\|^2$ and bound its differences similarly. By Assumption A2, if $\|\nabla g_k\|$ converged to a positive value, the sum would diverge, yielding a contradiction. Therefore, the limit must be zero: $\theta_k \to 0$ and $\nabla g_k \to 0$ almost surely. $\square$

Appendix C
Additional Experiment Details

The code is available at https://anonymous.4open.science/r/spectral-radius/. The algorithm is written in Python, using PyTorch [3] and TorchVision [4]. Forest cover-type and USPS experiments are run on a 3.1 GHz Dual-Core Intel Core i5 processor with 16 GB 2133 MHz LPDDR3 memory. Chest X-ray experiments are run on an Intel Xeon CPU E5-2650 v4 at 2.20 GHz with an NVIDIA Tesla K40c GPU.

A. Data Sets

The forest cover-type data [5] use cartographic data to predict the tree species (as determined by the United States Forest Service) of a 30×30-meter cell. These cartographic data include elevation, aspect, slope, distance to surface water features, distance to roadways, hill-shade index at three times of day, distance to wildfire ignition points, wilderness area designation, and soil type. Seven major tree species are included: spruce/fir, lodgepole pine, Ponderosa pine, cottonwood/willow, aspen, Douglas-fir, and krummholz. In total, 581,012 samples are included, which we split 64%/16%/20% into training, validation, and test data sets.

The USPS digits data [6] include 16×16-pixel grayscale images from scanned envelopes to identify the digit, 0–9, to which each image corresponds. These data are already split into 7,291 training and 2,007 test images. We take $1/7$ of the training set as validation.

The chest X-ray data [7] contain 1024×1024-pixel color images of patients' chest X-rays to identify which of fourteen lung diseases each patient has. This is a multilabel problem; patients can have none, one, or multiple of these conditions. A total of 112,120 patient images are included, taken between 1992 and 2015, which we split 70%/10%/20% into training, validation, and test data sets.

B. Generalization Tests

For forest cover-type data, we weight the test subjects to shift the mean of a feature or multiple features. Since we normalize the data, the weight of each test subject is determined by the ratio of the normal probability density function value with and without the shift. This shift adds a slight bias to the test set that is not in the original data set. We first use this shift method to increase the mean of each feature value by 0.1, compare the accuracy of our trained models, and find that certain features are problematic. Upon further examination, this is because these features are binary factors with rare classes (so our weighting of subjects emphasizes a few of them). We then shift each feature (except the problematic features) by a random normal amount (with mean 0 and standard deviation 0.05), compare the accuracy of our trained models, and repeat this procedure one thousand times.

For USPS handwritten-digits data, we augment the test set: Augmented Test 1 uses random crops (with padding) of up to one pixel and random rotations of up to $15°$. Augmented Test 2 uses crops of up to two pixels and rotations of up to $30°$. We do not augment our training set while learning our models, as a similar augmentation would yield comparable training and test sets.

For the conditional GAN examples, we modify Linder-Norén's [8] implementation and generate 10,000 images from the trained generator model. We train the GAN model with a batch size of 64, a cosine-annealing learning rate (initially 10^{-4}), $\beta_1 = 0.5$, and $\beta_2 = 0.999$. We randomly smooth the labels to be uniform between 0.0 and 0.3 for generated samples and 0.7 and 1.0 for true samples. We also swap the labels on 1% of batches, chosen at random. We modify Chhabra's [9] implementation similarly and generate 10,000 images from the trained generator model.

For the chest X-ray data, we compare performance on two similar data sets, CheXpert [10] and MIMIC-CXR [11]. For this comparison, we consider only the six conditions common to the three data sets: atelectasis, cardiomegaly, consolidation, edema, pneumonia, and pneumothorax. Additionally, we ignore uncertain labels in the CheXpert and MIMIC-CXR classes. We keep the assigned training and validation data sets separate for each data set, as there appear to be differences in labeling. In particular, the CheXpert validation set is fully labeled, while the training set contains uncertain and missing labels. Although these are labeled "training" and "validation" sets, we use them solely as test sets. CheXpert contains 234 validation and 223,415 training images from Stanford Hospital, taken between 2002 and 2017. MIMIC-CXR contains 2,732 validation and 369,188 training images from Beth Israel Deaconess Medical Center, taken between 2011 and 2016.

We measure the spectral radius ρ of each model on the full training set. We use $\epsilon = 10^{-3}$ and a maximum of 1,000 power iterations, except for the chest X-ray models, where we use $\epsilon = 0.1$ and a maximum of 100 power iterations. These values allow the algorithm to find an accurate eigenvalue within a reasonable run time.

C. Implementation

For forest cover-type models, we train using stochastic gradient descent, with a learning rate of $0.5/\text{epoch}$, a batch size of 128, and a maximum of 100 epochs. For models with batch sizes 32 and 64, we use a $0.1/\text{epoch}$ learning rate instead. The network uses three hidden layers with 20 hidden nodes in each layer. The learning rate and maximum number of epochs allow our algorithm to converge to an accurate model. We experiment with other feed-forward networks, learning rates, and optimizers but find that this structure works best. The corresponding batch-size experiments are reported in the main paper.

For USPS models, we use the Adam optimizer with a learning rate of 10^{-3}, a batch size of 128, and a maximum of 100 epochs. The network takes an input image and processes it through the following layers in order: convolution to 8 channels, pooling, convolution to 16 channels, pooling, convolution to 32 channels, pooling, a fully connected layer with 64 nodes, a fully connected layer with 10 nodes, and softmax. All convolutions have kernel size 3, stride 1, and padding 1; all pools are max-pooling layers with kernel size 2 and stride 2. ReLUs connect the various layers. The MNIST images are resized to 16×16 to match the model's input size. We similarly experiment with feed-forward and other convolutional networks, learning rates, and optimizers but find that this structure performs best.

For the chest X-ray models, we resize the images to 256×256 and then crop them to 224×224. We use the Adam optimizer with a learning rate of 10^{-5}, a batch size of 4, a maximum of 100 power iterations, random initialization of each power iteration, and gradient clipping (at a magnitude of 100). The crops and small batch size are necessary to prevent the GPUs on our server from running out of memory (12 GB). Using a CheXNet model [12], trained according to its authors' methodology, as the initialization, we train for an additional epoch, except for the asymmetric-valley model. We also try five epochs; however, the first epoch has the highest validation mean AUC in each case.

The entropy-SGD models are trained with a learning rate of 0.1 (except on chest X-ray data, where 0.001 is used), a momentum of 0.9, and no dampening or weight decay. The K-FAC models are trained using Wang's [13] implementation, with a learning rate of 0.001 (except on chest X-ray data, where 10^{-7} is used) and a momentum of 0.9.

The asymmetric-valley models are trained with an initial learning rate of 0.5 for 250 epochs (iterations 161–200 utilizing SWA). For chest X-ray data, we start at the SWA step, using CheXNet initialization [12].

The forest cover-type LOBPCG model is trained with regularization parameters $\mu = 0.0028$ and $K = 1$, update frequency $b = 4$, and learning rate $\tilde{\alpha}(j) = \exp(-4j - 2)$. The USPS LOBPCG model is trained with regularization parameters $\mu = 0.005$ and $K = 0$, update frequency $b = 4$, and learning rate $\tilde{\alpha}(j) = \exp(-4j)$.

Appendix D
Constructed Data Sets

During analysis of the performance of GAN1, we compute the minimum distance (L_2 norm) between each image in the GAN1 data set and the images in the USPS test data. We notice that the GAN1 images are distributed differently, relative to the USPS test data, compared with the other data sets. In particular, Fig. 1 shows that the augmented data sets have a bell-shaped distribution of such distances, while GAN1 has an abnormal distribution. We construct a data set, Const1, from the augmented test data sets using the following procedure:

1) Split the data with respect to distances into integer bins $[0,1), [1,2), \ldots, [17,18)$.
2) Uniformly at random, select five bins from which to draw zero images.
3) For the remaining bins, select one of the two augmented test data sets uniformly at random. Add all images from the selected data set in the bin's distance range to the constructed data set.

This procedure creates a constructed data set intended to emulate the abnormal distribution of the GAN1 data. We repeat this process with the maximum cosine similarity between images and observe similar distributional abnormalities in the GAN1 data (Fig. 2). We construct Const2 from the augmented data set using a similar procedure, but with bins $[0.5, 0.525), [0.525, 0.55), \ldots, [0.975, 1.0)$.

APPENDIX E
GRAD-CAM

We use Gildenblat's [14] Grad-CAM implementation to highlight which areas of the chest X-rays are important to our best regularized model ($\mu = 10^{-4}$ and $\alpha = 10^{-6}$) and the predictions of the baseline models. We compute the Jaccard index of the top 10% of pixels in each Grad-CAM image to compare which regions are important to each model.

In contrast, the three models with higher spectral radius—unregularized, K-FAC, and asymmetric valley—have a greater performance drop on the transfer-learning data sets and less overlap in their explanations. These models have mean Jaccard scores of 0.202–0.306 on the CheXpert validation data, lower than the spectral-radius-regularization and entropy-SGD scores of 0.342 and 0.351. The high-spectral-radius models have scores of 0.199–0.249 on MIMIC-CXR validation data, while the low-spectral-radius models have scores of 0.330 and 0.316.

Table I shows that the two models with the lowest spectral radius ρ, our regularized model and entropy-SGD, have the highest overlap in explanations. These models highlight similar areas of the chest X-rays as important in making predictions. The Jaccard scores of their overlap are over 0.5 on the CheXpert and MIMIC-CXR validation data, the highest of any pair of models. Since these models also perform best on these transfer-learning data, the evidence suggests that models with low spectral radius generalize better in both their explanations and predictions. The three models with higher spectral radii have a larger dip in performance and less overlap in their explanations.

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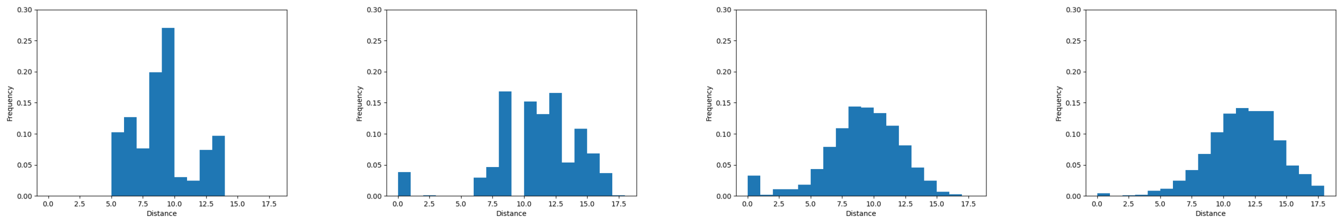

Fig. 1. Histogram of Euclidean distance from each USPS test image for GAN1, GAN2, Augmented Test 1, and Augmented Test 2.

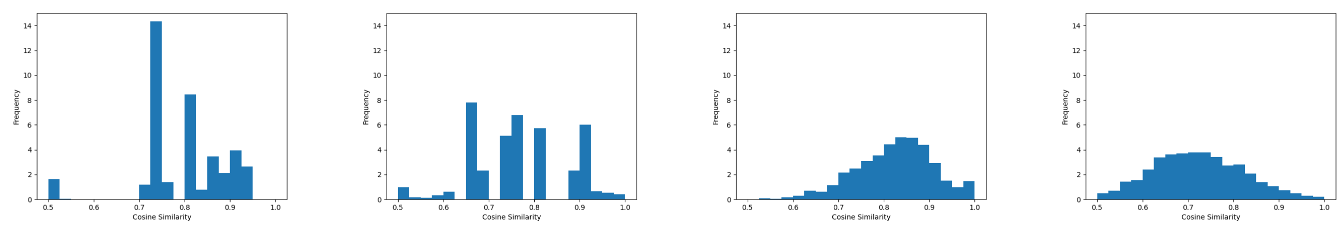

Fig. 2. Histogram of cosine similarity between each USPS test image for GAN1, GAN2, Augmented Test 1, and Augmented Test 2.

TABLE I

JACCARD-SCORE OVERLAP ON THE CHEXPERT AND MIMIC-CXR VALIDATION DATA SETS. THE LOW-SPECTRAL-RADIUS MODELS SHOW THE GREATEST EXPLANATORY OVERLAP.

(a) CheXpert validation data set

Model	ρ	SpecRad	EntropySGD	UnReg	KFAC	AsymValley	Mean	PerfDrop
SpecRad	38.92	1.000	0.508	0.192	0.294	0.374	0.342	-4.15%
EntropySGD	41.11	0.508	1.000	0.181	0.280	0.435	0.351	-3.09%
UnReg	68.29	0.192	0.181	1.000	0.283	0.152	0.202	-10.30%
KFAC	84.65	0.294	0.280	0.283	1.000	0.261	0.280	-13.16%
AsymValley	1198.69	0.374	0.435	0.152	0.261	1.000	0.306	-10.06%

(b) MIMIC-CXR validation data set

Model	ρ	SpecRad	EntropySGD	UnReg	KFAC	AsymValley	Mean	PerfDrop
SpecRad	38.92	1.000	0.504	0.216	0.285	0.313	0.330	-8.10%
EntropySGD	41.11	0.504	1.000	0.172	0.229	0.358	0.316	-9.21%
UnReg	68.29	0.216	0.172	1.000	0.272	0.134	0.199	-10.80%
KFAC	84.65	0.285	0.229	0.272	1.000	0.192	0.245	-10.71%
AsymValley	1198.69	0.313	0.358	0.134	0.192	1.000	0.249	-10.84%